JU-6 Pi

6-VOICE POLYPHONIC SYNTHESIZER · RASPBERRY PI 5

OWNER'S MANUAL



Software version 1.2.0 · Played from an AKAI MPK mini Plus over USB · Audio via HDMI or Raspberry Pi DAC+

JU-6 Pi is an independent hobby project inspired by the Juno-106. It is not affiliated with or endorsed by Roland Corporation.

Contents

1 Introduction

2 Important Notes

3 Panel Descriptions

4 Connections

5 Power-up and Status LED

6 Playing

7 Selecting Patches

8 Knobs and Takeover

9 Creating Sounds

10 Writing (Saving) Patches

11 Audio Output and Latency

12 Settings File (ju6.conf)

13 MIDI Implementation

14 Parameter Chart

15 Patch List

16 Troubleshooting

17 Specifications

18 Sources and Credits

NOTE Everything in this manual about controls, MIDI numbers, patch memory, audio outputs and latency was taken from the JU-6 Pi v1.2.0 source code and README (repository `OleBee/juno106-pi`). Items that could not be confirmed are marked **NOT VERIFIED**.

1 Introduction

Thank you for building the **JU-6 Pi**. It turns a Raspberry Pi 5 (8 GB) into a stand-alone, six-voice polyphonic synthesizer in the spirit of the classic 1984 Juno-106. There is no screen, no network and no setup: write the SD card, plug in an AKAI MPK mini Plus and an audio output, switch on and play.

The sound is generated in real time by the **Ultramaster KR-106** DSP engine (GPL-3), a component-level model of the Juno-106 calibrated against measurements of real hardware. JU-6 Pi wraps that engine in its own headless real-time program (`ju6pi`, C++17, direct ALSA, no JACK or PipeWire). No ROM dumps or samples are used — every sound is computed.

Main features

- **6 voices** with Juno-106 style voice assignment (Poly I), plus Poly II and Unison.
- **DCO** with pulse (PWM from LFO, manual or ENV), sawtooth, sub-oscillator (square, one octave down), noise; RANGE 16' / 8' / 4'.
- **VCF** — 4-pole resonant low-pass that self-oscillates, with ENV (±polarity), LFO and keyboard follow; 2× or 4× oversampled.
- **HPF** with four positions; position 0 is the 106's bass boost.
- **VCA** switchable between ENV and GATE, with LEVEL. One **ADSR envelope** and a triangle **LFO** with RATE and DELAY.
- **Chorus** I, II and I+II, modelled as a bucket-brigade (BBD) circuit with companding and noise.
- **128 patch memories** (A11–B88) pre-loaded with the factory sounds, stored in genuine Juno-106 SysEx format.
- Two **panel sides** on the MPK pads and knobs cover the Juno-106 sound parameters. Pad 6 and pad 2 select the side, and their LEDs show which side is active.
- Read-only system on the SD card — you may pull the power at any time.

To get the most out of the instrument, please read this manual from beginning to end. Keep it handy for later reference.

The DSP engine follows the Juno-106 closely, but JU-6 Pi is *not* a Roland product. "Roland" and "Juno" are trademarks of their respective owners and are used here only to describe what the synthesizer emulates.

2 Important Notes

Power supply

- Use the official Raspberry Pi **27 W USB-C** power supply (recommended in the project README). The Pi also powers the MPK mini Plus through its USB port.
- The root file system is read-only (overlayroot). The only thing the synthesizer writes to is the **JU6DATA** partition, and it writes atomically (new file, fsync, rename). It is therefore safe to switch off simply by pulling the power — there is no shut-down procedure.

Placement and handling

- Use the Pi with adequate cooling (the project is built around a Pi 5 with the **Active Cooler**; the audio thread keeps one CPU core busy and the CPU is held at full clock).
- When fitting a DAC+ board, check that no GPIO pin is shifted by one row **before** applying power.
- Do not connect or disconnect the DAC+ board while the Pi is powered.

Listening level

- Turn down the amplifier, TV or monitors before switching on. Two short **ready tones** are played about 10–15 seconds after power-up.
- The VCF can self-oscillate at high RES settings and the HPF bass boost (position 0) adds low end. Keep the volume moderate while experimenting.

Memory backup

- Your patches are stored in `ju6/patches.syx` and `ju6/names.txt` on the JU6DATA partition. Put the SD card in a computer from time to time and copy these files to a safe place.
- Deleting `patches.syx` and `names.txt` restores the factory sounds.

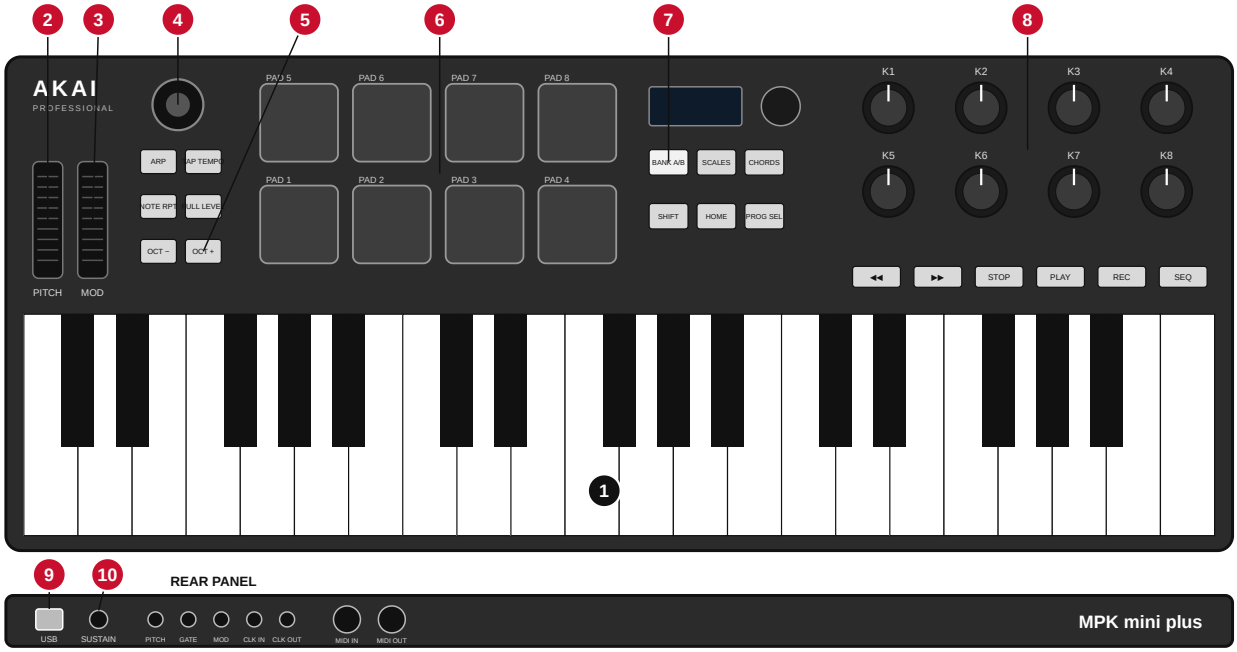
What is not on board

For a clean, fast boot the image has Wi-Fi, Bluetooth, SSH, NetworkManager, cloud-init, avahi, cron, apt timers and swap switched off. The system log is kept in RAM only. All configuration is done by editing a text file on the SD card (see section 12).

> **CAUTION** This manual describes software version **1.2.0**. The panel is two sides (section 3). Sounds are chosen one patch at a time with pad 5 and pad 1. The panel does not write patches and has no transpose control. DAC+ support arrived in 1.1.0.

3 Panel Descriptions

JU-6 Pi has no panel of its own — the **AKAI MPK mini Plus** is its front panel. The numbers below refer to the drawing; only controls that JU-6 Pi uses are numbered.



AKAI MPK mini Plus, top and rear panel (drawing based on the Top/Rear Panel figures in AKAI's MPK mini Plus User Guide v1.2; simplified, not to scale). Note that pads 1–4 are the **lower** row and pads 5–8 the **upper** row.

1. **Keyboard** — 37 keys. Plays JU-6 Pi on every MIDI channel (omni). Velocity is ignored by default, like the original. Also used to set the key with TRANSPOSE (pad bank B, pad 8).
2. **Pitch wheel** — BENDER. Bends the DCO (default depth 0.25 ≈ ±2 semitones) and optionally the VCF. Depths are `bender_dco` and `bender_vcf` in `ju6.conf`.
3. **Modulation wheel** (CC 1) — MOD: LFO vibrato on the DCO. Depth follows how far the wheel is moved (scaled by `mod_depth`, default 0.5).
4. **X-Y joystick** — JU-6 Pi reacts to whatever the joystick sends: **Pitch Bend** acts as BENDER, **CC 1** acts as MOD. Other CC numbers are ignored unless listed in `mod_cc`. *NOT VERIFIED: which messages the joystick sends in the MPK's default program.*
5. **OCT − / OCT +** — handled inside the MPK; shifts the keyboard range. Press both to return to the centre octave.
6. **Pads 1–8** — the two panel sides below. Bank A and bank B send different notes, and both are the same panel. **Pad 6** (upper row) selects SIDE 1. **Pad 2** (lower row, directly under pad 6) selects SIDE 2. The pad for the active side stays lit strong red; the other stays off. **Pad 5** is the next sound and **pad 1** the previous sound.
7. **BANK A/B** — switches which notes the pads send. JU-6 Pi treats both banks as this panel, so the layout does not change. LEDs are sent for both banks' notes.
8. **Knobs K1–K8** (CC 70–77) — sound parameters. What each knob does depends on the selected side (below).
9. **USB port** — connect to any USB port on the Pi. Carries MIDI and powers the MPK.
10. **Sustain input** — a sustain pedal (CC 64) works as HOLD.

NOTE The other MPK functions (arpeggiator, scales, chords, note repeat, sequencer, transport buttons, CV/Gate, 5-pin MIDI) are run by the MPK itself. They are not used or tested by JU-6 Pi; any notes they produce over USB are simply played. OCT −/+ still shift the keyboard inside the MPK. DCO RANGE is stored in each patch and is not on the panel. The modulation wheel is still LFO vibrato on the DCO.

The two sides

JU-6 Pi starts on **SIDE 1**. On pad bank B, press **pad 6** for SIDE 1 or **pad 2** for SIDE 2. Both presses work from either side. The green ACT LED on the Pi blinks once for SIDE 1 and twice for SIDE 2.

The page pads themselves show which side is active. While SIDE 1 is selected, pad 6 stays lit strong red and pad 2 stays off. While SIDE 2 is selected, pad 2 stays lit strong red and pad 6 stays off. Switching sides updates these two LEDs immediately, together with the LEDs of the pads whose job changes. The same two notes are sent for bank A and bank B, so the active side stays visible after you press BANK A/B.

> **NOTE** The sketch names pad 2 and pad 6 as the page switches and draws SIDE 1 above SIDE 2. It does not spell out which pad is which. Pad 6 is the upper pad of the pair and selects the upper side; pad 2, directly beneath it, selects SIDE 2.

SIDE 1

lower row = pads 1–4

PAD 5	PAD 6	PAD 7	PAD 8
NEXT sound +1	**SIDE 1** lit when active	**LEVEL +** VCA, step 8	**CHORUS** off · I · II · I+II
PREVIOUS sound −1	**SIDE 2** off on side 1	**LEVEL −** VCA, step 8	**ENV / GATE** red = ENV

SIDE 2

same next/previous and page pads

PAD 5	PAD 6	PAD 7	PAD 8
NEXT sound +1	**SIDE 1** off on side 2	**ENV POL** red = on	**LFO / MAN** red = LFO
PREVIOUS sound −1	**SIDE 2** lit when active	**PULSE** red = on	**SAW** red = on

A **sound** is one patch memory: slot 0 is A11, slot 7 is A18, slot 8 is A21, slot 63 is A88, slot 64 is B11 and slot 127 is B88. Next adds one and wraps from B88 to A11. Previous subtracts one and wraps from A11 to B88. The step crosses group and bank boundaries. The Pi LED does not blink for a sound step; it blinks only when the side changes.

SIDE 1 · K1–K4 top, K5–K8 bottom

K1	K2	K3	K4
ENV A	ENV D	ENV S	ENV R
K5	**K6**	**K7**	**K8**
DCO SUB	DCO NOISE	VCF FREQ	VCF RES

SIDE 2 · K1–K4 top, K5–K8 bottom

K1	K2	K3	K4
DCO LFO	DCO PWM	LFO RATE	LFO DELAY
K5	**K6**	**K7**	**K8**
HPF 0·1·2·3	VCF ENV	VCF LFO	VCF KYBD

What the pad LEDs mean

Brightness is one red lamp in four steps: **off**, **weak red**, **red**, **strong red**. On a page change, every pad LED is set again for the side you just selected.

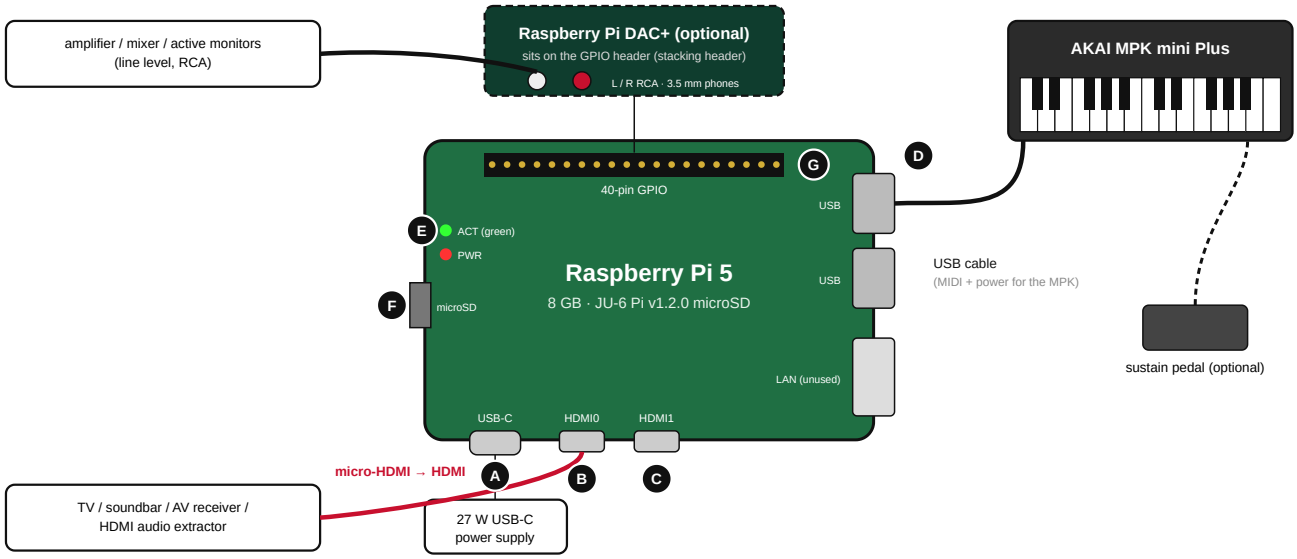
Pad	SIDE 1	SIDE 2
6 / 2	The active side is **strong red**. The other page pad is **off**. This pair is updated on every page change and is sent for both pad banks.	
5 and 1	**Next sound** (pad 5) and **previous sound** (pad 1). One patch per press, wrapping through all 128. The LEDs stay off. Both sides.	
7 and 3	**VCA LEVEL +** (pad 7) and **LEVEL −** (pad 3). Each press moves the level by 8; holding repeats. Both LEDs show the level: weak red at 0, strong red at 127, scaled in between.	Pad 7 = **VCF polarity**. Red = on (inverted), unlit = off (normal). Pad 3 = **pulse** on/off: red = on, unlit = off.

8 and 4	Pad 8 = **CHORUS**, each press cycles off → I → II → I+II → off. LED off / weak red / red / strong red. Pad 4 = **ENV / GATE**. Red = ENV, unlit = GATE.	Pad 8 = **PWM LFO / MAN**. Red = LFO, unlit = MAN. Pad 4 = **saw** on/off. Red = on, unlit = off. Pulse and saw are independent.

NOT VERIFIED *Pad LEDs are sent as Note On (pad channel, the pad's note, velocity 0 = off, 42 = weak red, 84 = red, 127 = strong red) to USB-MIDI ports whose name contains "MPK", never to a port named MIDI 2. AKAI's MPK mini Plus User Guide v1.2 only documents colours stored in the program, not live feedback, so this has* ***not been verified on the hardware****. If the pads do not respond, set* `pad_leds = off`*. The Pi's ACT LED still blinks the side number.* `pad_leds = any` *sends the same notes to every MIDI output except MIDI 2.*

NOTE A patch can store PWM from the envelope. The LFO/MAN pad only selects LFO or MAN: if the patch is on ENV the LED stays off, and the next press selects LFO. Loading the patch again brings ENV back. Voice mode, velocity, portamento, bender depths and MOD depth are not on the panel; they stay as set in `ju6.conf`. CC 7 is still the master volume, and a sustain pedal (CC 64) is still HOLD.

Raspberry Pi 5 (the sound module)



Simplified connection drawing (not to scale). The DAC+ board is optional.

A	**USB-C power** — 27 W supply recommended.
B	**HDMI0** (micro-HDMI, the port **nearest the USB-C power connector**) — primary HDMI audio output.
C	**HDMI1** — used automatically as a reserve if HDMI0 is not available.
D	**USB ports** — MPK mini Plus (any port). Other class-compliant USB-MIDI devices are connected automatically and can be hot-plugged. A USB audio interface can be used as an output.
E	**ACT LED (green)** — the synthesizer's status display (section 5).
F	**microSD slot** — the JU-6 Pi card (system + JU6DATA partition).
G	**40-pin GPIO header** — mount the optional Raspberry Pi DAC+ here (on a stacking header if you use the Active Cooler).

Raspberry Pi DAC+ (optional)

The **Raspberry Pi DAC+** (formerly IQaudIO DAC+) is a sound card that sits on the GPIO pins. It has a TI PCM5122 converter, two RCA line outputs (0–2 V RMS) and a 3.5 mm headphone output carrying the same stereo signal. It has no input. Older black IQaudIO DAC+ boards work with the same overlay.

> **NOTE** **HiFiBerry DAC+:** from v1.1.1 the ALSA card of a HiFiBerry DAC+ (`snd_rpi_hifiberry_dacplus`) is recognised as a DAC+ output. The image still loads only the `rpi-dacplus` overlay, so for a HiFiBerry board change the line `dtoverlay=rpi-dacplus` in `config.txt` on the boot partition to `dtoverlay=hifiberry-dacplus`. NOT VERIFIED on hardware. This manual describes the Raspberry Pi DAC+.

4 Connections

CAUTION To prevent damage, turn the volume of your amplifier or TV all the way down and leave the Pi unpowered while making connections.

Preparing the SD card

1. Download `ju6pi-1.2.0.img.xz` from the project's Releases page. If it is split into `.part-aa`, `.part-ab` … join the parts first (`cat ju6pi-1.2.0.img.xz.part-* > ju6pi-1.2.0.img.xz`) and check it with `sha256sum -c ju6pi-1.2.0.img.xz.sha256`.
2. Write it to a microSD card (8 GB or more). **Raspberry Pi Imager:** Choose Device → Raspberry Pi 5; Choose OS → Use Custom → the .img.xz file; Choose Storage → the card. When asked about OS customisation, answer **No**. **balenaEtcher:** Flash from file → Select target → Flash!
3. Insert the card into the Pi.

Connecting the equipment

1. **Audio, with DAC+:** connect the RCA outputs to an amplifier, mixer or active monitors. The synthesizer uses the board automatically when it is fitted.
2. **Audio, without DAC+:** connect **HDMI0** — the micro-HDMI port nearest the USB-C power connector — to a TV, soundbar, AV receiver or HDMI audio extractor. No screen is needed; the image supplies its own EDID and forces stereo PCM at 48 kHz on both ports.
3. **Controller:** connect the AKAI MPK mini Plus to any USB port on the Pi.
4. Optionally connect a sustain pedal to the MPK's SUSTAIN input.
5. Connect the USB-C power supply last.

Mounting the DAC+ in a KKSB case with Active Cooler

1. Fit the Active Cooler and connect its cable to the FAN connector.
2. Press the 40-pin stacking header supplied with the case onto the GPIO pins; it lifts the DAC above the cooler.
3. Fit the DAC+ on top with its RCA sockets towards the case openings, and secure it with the supplied standoffs and screws.
4. Check that no pin is offset by a row before switching on.

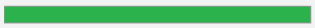
NOTE An HDMI audio extractor typically adds 0–2 ms. TVs and AV receivers can add 20–100 ms of their own processing — choose a **Game**, **Direct** or **Pure** mode on such devices.

5 Power-up and Status LED

1. Turn the amplifier or TV volume down.
2. Connect the power supply. The green ACT LED blinks slowly while the system starts.
3. After about **10–15 seconds** two short **rising** ready tones (A–E) are heard and the LED lights steadily. The synthesizer is now ready.
4. Raise the volume gradually while playing.

The last selected patch is recalled at power-up (it is remembered in `ju6/state.txt`; on the very first start, `start_patch` decides). The program is a systemd service that restarts automatically within one second if it ever stops.

Status LED (green ACT LED on the Pi)

	Slow blinking (1 Hz)	Starting up, or waiting for an audio output.
	**Steady on**	Ready to play.
	**On, two short gaps every 3 s**	Ready, but on a **reserve output**: `audio_output = dacplus` is set and the DAC+ was not found.
	**On, one short gap every 2 s**	Ready, but **no MIDI device** found — check the MPK's USB cable.
	**Rapid blinks**	Side or group changed (one blink = SIDE 1, two blinks = SIDE 2, or the group number 1–8), or a patch written (5 blinks).

If the ready tones are **falling** (E–A) instead of rising, the synthesizer is playing on a reserve output because the DAC+ you asked for was not found (see section 11).

Checking without a screen

Every few seconds the synthesizer writes `ju6/status.txt` on the JU6DATA partition: software version, chosen audio output, sound cards found, buffer size and latency, VCF oversampling, number of MIDI devices and underrun count. Put the SD card in a computer to read it. With DAC+ it should contain lines like:

```
lyd: Raspberry Pi DAC+ (RCA/hodetelefon) (plughw:CARD=RPiDAC,DEV=0)
lydkort funnet: vc4hdmi0, vc4hdmi1, RPiDAC
```

The status file is written in Norwegian: *lyd* = audio, *lydkort funnet* = sound cards found, *ADVARSEL* = warning.

6 Playing

Keyboard

The keyboard plays on all MIDI channels (`key_channel = omni`); only the pad notes on the pad channel are reserved. **Velocity is off** by default, as on the original; switch it on with `velocity = on`. Use the MPK's OCT −/+ buttons to reach higher and lower octaves. Those buttons do not change DCO RANGE. There is no transpose control on the panel: a key always plays.

BENDER (pitch wheel / joystick X)

Pitch Bend bends the DCO pitch by up to ±2 semitones (default `bender_dco = 0.25`) and, if set, the VCF cutoff (default `bender_vcf = 0`). The depths are set in `ju6.conf`.

MOD (mod wheel / CC 1)

MOD applies LFO vibrato to the DCO. Unlike the 106's spring-loaded lever with fixed depth, MOD here is **continuous**: the depth follows how far the wheel or joystick is pushed, scaled by `mod_depth` (default 0.5). The LFO runs in AUTO mode (it fades in after a key, using LFO DELAY on SIDE 2).

HOLD

A sustain pedal (CC 64) holds the notes while pressed. There is no HOLD pad on the two sides; CC 120 or 123 silences hung notes.

Voice modes

Mode	`voice_mode`	Description
Poly I (default)	1	Juno-106 voice assignment, as on the hardware.
Poly II	2	Alternative polyphonic assignment (an addition — not on the 106).
Unison	0	All voices on one note (an addition — not on the 106).

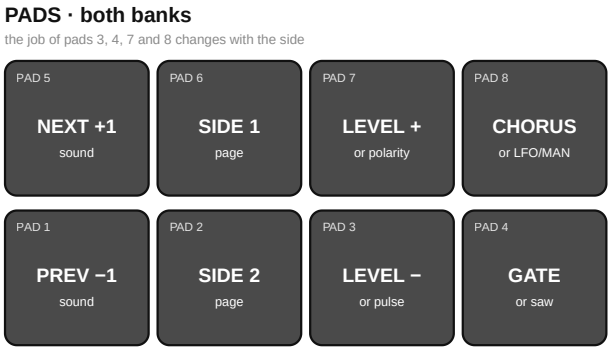
PORTAMENTO is not on the panel and starts at 0. Neither portamento, Unison nor velocity exist on the original instrument. They are not stored in a patch.

Other controls

MIDI CC 7 sets the master volume. CC 120 and 123 silence all notes; CC 121 resets bender and MOD.

7 Selecting Patches

As on the 106, the 128 memories are organised as two **banks** (A and B), each with eight **groups** of eight **patches**: A11 … A88 and B11 … B88. The first digit is the group, the second the patch number.



BOTH PAD BANKS

Bank A and bank B do the same panel.

Pad 6 selects SIDE 1. Pad 2 selects SIDE 2.

The active side's pad stays lit;
the other page pad stays off.

Pad 5 = next sound. Pad 1 = previous.
One sound = one patch, slots 0–127.

Both MPK pad banks are this panel. Pad 5 steps forward one sound. Pad 1 steps back one sound. Pad 6 stays lit on SIDE 1; pad 2 stays lit on SIDE 2.

What a sound is

A sound is **one patch**, not a group of eight. The 128 memories stay numbered as on the 106 — two banks, eight groups, eight patches — but next and previous move a single step:

Slot	Name	Next	Previous
0	A11	A12	B88
7	A18	A21	A17
63	A88	B11	A87
127	B88	A11	B87

The step crosses from patch 8 of a group into patch 1 of the next group, and from A88 into B11. It loads the stored patch and keeps the master volume. Knob edits that were not part of that memory are replaced. The Pi's green LED does not blink for this; it blinks only for a side change (one blink = SIDE 1, two = SIDE 2).

Example: from A11, press NEXT seven times to reach A18, once more to reach A21. Thirty-nine presses from A11 reach A48 Synth Bass I. PREVIOUS from A11 lands on B88.

Program Change

MIDI Program Change 0–127 selects A11 … B88 directly (0 = A11, 63 = A88, 64 = B11, 127 = B88). The MPK can send program changes from its pads (pad Play Mode "ProgCh" in the MPK's program editor, per AKAI's guide).

SysEx

A Juno-106 patch dump (APR, `F0 41 30 0n …`) or single-parameter message (IPR, `F0 41 32 0n …`) received over MIDI is loaded into the current sound, just as on a 106. This lets you audition sounds from Juno-106 editors and librarians.

Master volume across patches

Selecting a patch changes every sound parameter but keeps the master volume where it was. Pad LEDs are refreshed to match the new sound and the side you are on, including pad 2 and pad 6.

8 Knobs and Takeover

The eight knobs send **CC 70–77** (K1–K4 on the top row, K5–K8 on the bottom). Their function is the side selected with pad 6 or pad 2 (section 3). SIDE 1 is active after power-up. The drawings of both sides are in section 3.

The one switched knob

Most knobs are continuous, 0–127. Two are not:

Control	How it moves
SIDE 1, K5 SUB	0 = sub off; 1–127 = sub level
SIDE 2, K5 HPF	0–31 position 0 (bass boost) · 32–63 position 1 (flat) · 64–95 position 2 · 96–127 position 3

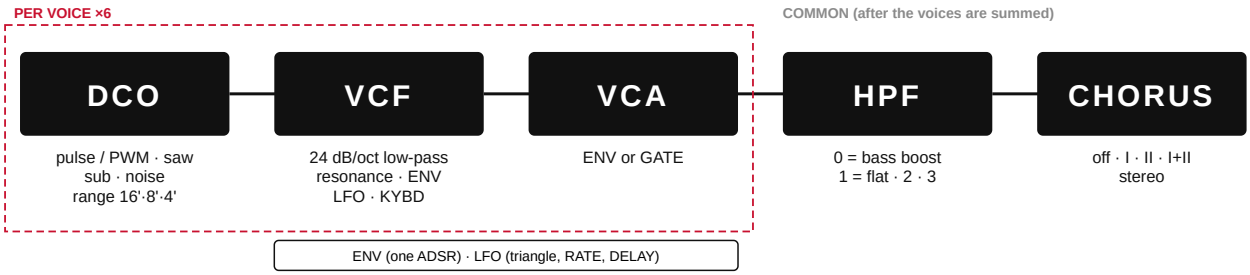
Chorus, pulse, saw, PWM source, VCA mode and envelope polarity are pads, not knobs (section 3). DCO RANGE is stored in the patch and has no panel control.

Knob behaviour

By default the knobs **jump**: the parameter takes the knob's value as soon as the knob is moved (`knob_takeover = jump`). With `knob_takeover = pickup` a knob does nothing until it passes the current value (soft takeover; this resets on every side or patch change). With `knob_mode = relative` JU-6 Pi reads the knobs as relative (two's complement: 1–63 = up, 65–127 = down). *NOT VERIFIED: that this matches the MPK mini Plus's own "Relative" knob mode.*

9 Creating Sounds

Each of the six voices has its own DCO, VCF and VCA. The voices are summed and pass through the common HPF and chorus. A single envelope (ENV) and a single LFO control all voices.



Section	What it does	Where
DCO	The sound source. Pulse (width set by PWM, from LFO or set manually), sawtooth, sub-oscillator one octave down, and noise are mixed. RANGE is the octave stored in the patch. DCO LFO adds vibrato.	SIDE 2, plus SUB and NOISE on SIDE 1
VCF	Low-pass filter, 24 dB/octave. FREQ = cutoff, RES = resonance (self-oscillates near maximum), ENV = how much the envelope opens the filter, LFO = filter wobble, KYBD = cutoff follows the keyboard. Polarity is the SIDE 2 pad (red = inverted).	Both sides
VCA	Loudness. ENV: follows the ADSR. GATE: full level while a key is held. LEVEL sets the patch volume (about −10 … +10 dB), in steps of 8 from the LEVEL pads.	SIDE 1 pads
ENV	One ADSR envelope with Juno-106 curves: A = attack, D = decay, S = sustain level, R = release.	SIDE 1, K1–K4
LFO	Triangle wave. RATE = speed, DELAY = fade-in time after a key is pressed.	SIDE 2, K3–K4
HPF	Position 0 = bass boost (about +9.4 dB low shelf), 1 = flat, 2 ≈ 236 Hz, 3 ≈ 754 Hz high-pass.	SIDE 2, K5
Chorus	I = slow, wide; II = faster, deeper; I+II = a third, distinct mode (fast, shallow, vibrato-like). The pad cycles off → I → II → I+II.	SIDE 1, pad 7

How to use the tutorials

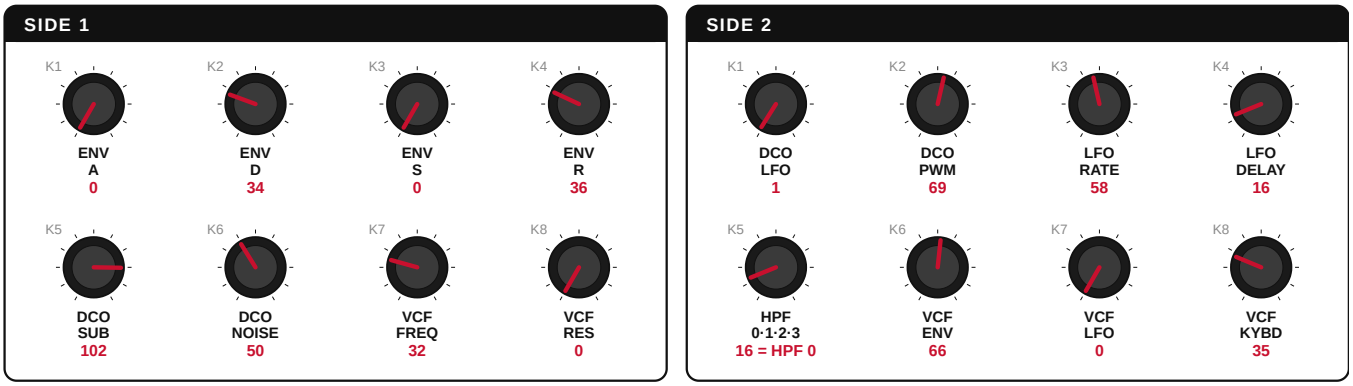
There is no "initialise patch" function, so every tutorial lists the knobs on both sides and the pad states. Select the side with pad 6 or pad 2, then turn each knob until it sends the target value (0–127). The patches are re-creations of factory sounds, so you can recall the original from bank A at any time to compare — but note that this replaces what you have dialled in.

> **NOTE** Start from a patch in a group you don't mind experimenting in. Because the knobs jump by default, each knob must actually be turned at least a little so it sends its value.

Tutorial 1 — Synth bass (A48 Synth Bass I)

A punchy, mono-style bass made from a narrow pulse, a strong sub-oscillator and a short filter envelope.

1. Recall A48 first (or step to it) so DCO RANGE is already 16'. SIDE 2: pad 3 pulse on (red), pad 4 saw off, pad 8 LFO/MAN unlit (MAN). Knob PWM (K2) = 69. DCO LFO (K1) = 1.
2. SIDE 1: sub-oscillator K5 = 102 for weight, and a touch of NOISE (K6 = 50) for attack. Filter closed: FREQ 32, RES 0.
3. SIDE 2: let the envelope open the filter (VCF ENV 66). KYBD 35. HPF position 0 (K5 = 16) for the bass boost. LFO RATE 58, DELAY 16, VCF LFO 0.
4. SIDE 1: envelope A = 0, D = 34, S = 0, R = 36 — a plucked shape with no sustain. CHORUS (pad 8) until the LED is off. ENV/GATE (pad 4) until it is red (ENV). LEVEL pads (7 and 3) toward 75 (steps of 8; 72 or 80 is close).
5. Play low on the keyboard. Try more RES or VCF ENV for a squelchier bass.

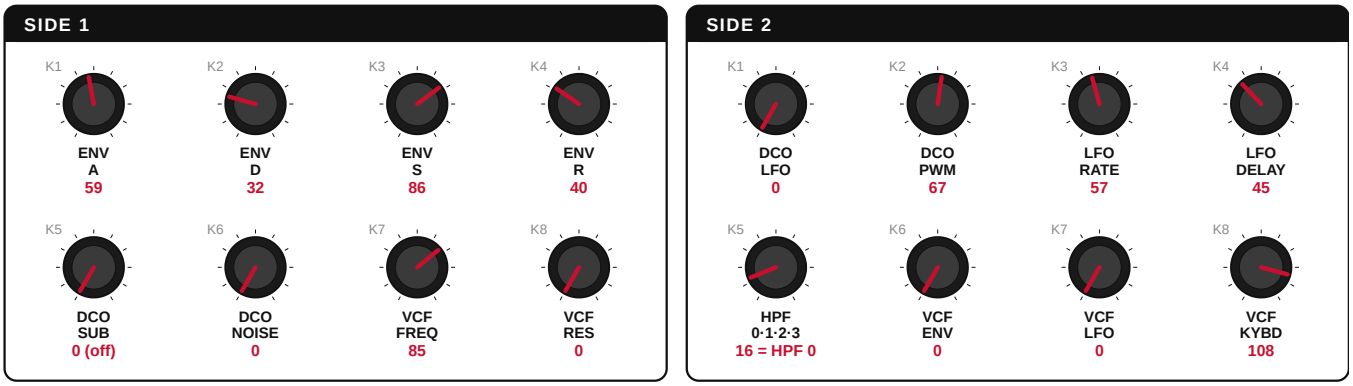


Pads for **A48 Synth Bass I**: pulse/saw pulse · PWM MAN · CHORUS off · VCA ENV · ENV polarity + · LEVEL 75 · RANGE stays 16' (stored in the patch, not on a pad). LEVEL moves in steps of 8, so aim for the nearest step to the factory value. Knob numbers are 0–127, computed from `src/factory_bank.h` with the same scaling the synth uses when it loads a patch.

Tutorial 2 — Strings (B11 Strings)

A classic ensemble: pulse and saw together, slowly swept by the LFO, softened by the envelope and widened by chorus II.

1. Recall B11 first so RANGE is already 8'. SIDE 2: pad 3 and pad 4 both on (pulse and saw, both red). Pad 8: LFO/MAN red (LFO), PWM knob 67 — the LFO sweeps the pulse width. VCF KYBD 108. HPF position 0 (K5 = 16). LFO RATE 57, DELAY 45, VCF LFO 0. VCF ENV 0.
2. SIDE 1: SUB 0 (off), NOISE 0. Open the filter fairly far (FREQ 85), RES 0 — the tone stays constant.
3. SIDE 1: slow envelope: A = 59, D = 32, S = 86, R = 40. CHORUS (pad 8) until the LED is red (II). Compare with weak red (I) and strong red (I+II).
4. ENV/GATE pad red (ENV). LEVEL pads toward 52.
5. Play chords. Raise ENV A for slower bowing, or ENV R for longer tails.

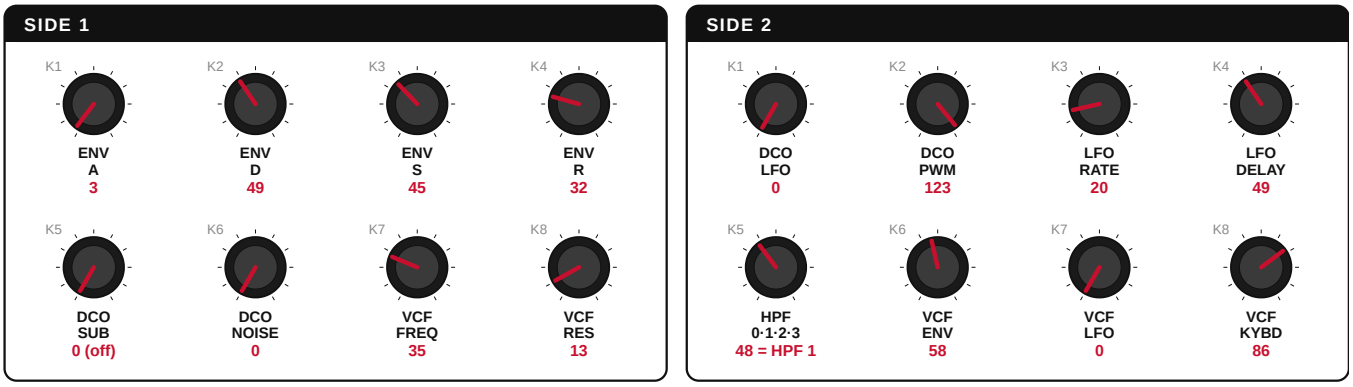


Pads for **B11 Strings**: pulse/saw pulse+saw · PWM LFO · CHORUS II · VCA ENV · ENV polarity + · LEVEL 52 · RANGE stays 8' (stored in the patch, not on a pad). LEVEL moves in steps of 8, so aim for the nearest step to the factory value. Knob numbers are 0–127, computed from `src/factory_bank.h` with the same scaling the synth uses when it loads a patch.

Tutorial 3 — Brass (A11 Brass)

Synth brass relies on a sawtooth and a filter that opens with the envelope and settles back, giving the 'blat' of a horn attack.

1. Recall A11 first so RANGE is already 16'. SIDE 2: pad 4 saw on (red), pad 3 pulse off, pad 8 unlit (MAN). The PWM knob has no audible effect while the pulse is off. VCF KYBD 86. VCF ENV 58 — the envelope does the work. LFO RATE 20, DELAY 49, VCF LFO 0. HPF position 1 = flat (K5 = 48).
2. SIDE 1: SUB 0, NOISE 0. FREQ 35, RES 13. Envelope A = 3, D = 49, S = 45, R = 32: fast attack, the filter falls back to a sustained level.
3. CHORUS (pad 8) until the LED is weak red (I). ENV/GATE (pad 4) red (ENV). LEVEL pads toward 108.
4. Use the mod wheel for vibrato on held notes. For a slower swell, raise ENV A to about 60.



Pads for **A11 Brass**: pulse/saw saw · PWM MAN · CHORUS I · VCA ENV · ENV polarity + · LEVEL 108 · RANGE stays 16' (stored in the patch, not on a pad). LEVEL moves in steps of 8, so aim for the nearest step to the factory value. Knob numbers are 0–127, computed from `src/factory_bank.h` with the same scaling the synth uses when it loads a patch.

10 Writing (Saving) Patches

The two-page panel does not write patches. Next and previous load the stored sound; they do not keep unsaved knob edits in the memory you just left. A file already on the card is still read at power-up. In 1.1.1 a held bank-A pad wrote the edited sound; that gesture is gone.

1. Edit the sound with the knobs and pads.
2. To hear a stored memory again, step to it with pad 5 or pad 1, or send Program Change.
3. Stepping away replaces the edit with the stored patch. The master volume stays where it was.

NOTE 1.1.1 wrote a patch when a bank-A pad was held. 1.2.0 does not. `patches.syx` on the card is still loaded if it is there; deleting it restores the factory sounds.

Patch files on the JU6DATA partition

File	Contents
`ju6/patches.syx`	All 128 memories as standard Juno-106 SysEx (APR) messages; created the first time you write. Can be opened in Juno-106 editors.
`ju6/names.txt`	One line per user patch: `A11<TAB>name[<TAB>ext]`. *ext* bit 0 = chorus I+II, bit 1 = PWM from ENV.
`ju6/fabrikk-A-B.syx`	Copy of factory banks A and B for reference.
`ju6/state.txt`	Last selected patch and buffer setting (automatic).
`ju6/status.txt`	What the synth found at the last start-up.
`ju6/ju6.conf`	Your settings (section 12).

Juno-106 format compatibility

The Juno-106 patch format has no room for chorus **I+II** or **PWM from ENV**. JU-6 Pi stores these in the extra column of `names.txt`, so they come back on JU-6 Pi; a real 106 or an editor sees the nearest valid setting (chorus I, PWM MAN).

Restoring factory sounds

Delete `patches.syx` and `names.txt` from the JU6DATA partition. Deleting `state.txt` resets the last-patch and buffer memory.

11 Audio Output and Latency

The output is chosen with `audio_output` in `ju6/ju6.conf`. Cards are found by name, not number, so ALSA's card order doesn't matter.

`audio_output`	Behaviour
`auto` (default)	Raspberry Pi DAC+ if fitted, otherwise HDMI0, then HDMI1, then a USB audio interface.
`dacplus`	Raspberry Pi DAC+. If it is missing, plays on HDMI0/HDMI1/USB but warns: falling ready tones, LED with two short gaps every 3 s, and an `ADVARSEL` (warning) line in `status.txt`.
`hdmi`	Only HDMI0, then HDMI1 — exactly as in 1.0.0. The DAC+ is ignored even if fitted.
`usb`	Only a USB audio interface.
`null`	No audio (testing).

At start-up the synth waits up to `dacplus_wait` = 8 s for the DAC driver when the board's HAT EEPROM reports a DAC or `audio_output = dacplus`. If it started on HDMI and the DAC+ appears within the first minute, it switches over and plays the ready tones again there. DAC+ output is 48 kHz, 32-bit; the mixer is set to 0 dB at start-up (2 V RMS). Lower it with `dacplus_volume_db` (0 or negative). HDMI output is 2-channel PCM at 48 kHz.

Buffer sizes (48 kHz)

The synth starts with the smallest buffer and moves up one step automatically after repeated underruns (3 or more within 30 s). The step is remembered for the next start.

Step	HDMI	HDMI buffer	DAC+	DAC+ buffer	USB	USB buffer
1 (start)	256 × 2	**10.7 ms**	128 × 2	**5.3 ms**	64 × 2	**2.7 ms**
2	256 × 3	16.0 ms	128 × 3	8.0 ms	128 × 2	5.3 ms
3	512 × 2	21.3 ms	256 × 2	10.7 ms	128 × 3	8.0 ms
4	512 × 3	32.0 ms	256 × 3	16.0 ms	256 × 2	10.7 ms
5	1024 × 2	42.7 ms	512 × 2	21.3 ms	256 × 3	16.0 ms
6	1024 × 3	64.0 ms	512 × 3	32.0 ms	512 × 2	21.3 ms

Period × number of periods, from `src/audio_out.h`. A fixed size can be forced with e.g. `buffer = 512x2`.

Key-to-sound latency

Estimated as USB-MIDI (≈1 ms) + one period of processing + the buffer + the receiving device:

Output (step 1)	Estimated latency out of the Pi
Raspberry Pi DAC+ (RCA)	≈ 8–10 ms
HDMI	≈ 12–17 ms, plus the receiver (extractor 0–2 ms; TV/AVR 20–100 ms)
USB audio interface (step 2)	≈ 7–10 ms total

NOT VERIFIED *These latency figures are **calculated, not measured** on hardware — the project README states that none of them has been measured yet. The actual output and buffer chosen are reported in `ju6/status.txt`.*

Real-time setup

The audio thread runs with SCHED_FIFO priority 80, locked memory, pinned to CPU 3, which is isolated from other processes (`isolcpus=3`). The CPU governor is fixed at *performance* and USB autosuspend is off so the controller never sleeps. VCF oversampling is chosen at start-up: 4× if the CPU manages it with a good margin (≥ 2.5× real time), otherwise 2×.

12 Settings File (ju6.conf)

All settings live in `ju6/ju6.conf` on the **JU6DATA** partition (FAT, 256 MB), which any computer can read. Every line is commented out; remove the `#` in front of a line and change the value. No recompiling is needed — the new settings apply at the next power-up.

Key	Default	Meaning
`audio_output`	auto	auto, dacplus, hdmi, usb or null (section 11)
`dacplus_volume_db`	0	DAC+ “Digital” mixer level in dB (0 or negative)
`dacplus_wait`	8	seconds to wait for the DAC+ at start-up
`audio_devices`	(not set)	legacy priority list (dacplus hdmi0 hdmi1 usb null); used only without audio_output
`buffer`	auto	auto, or fixed PERIODxCOUNT, e.g. 512x2
`oversample`	auto	VCF oversampling 2 or 4
`chime`	on	ready tones
`output_gain_db`	0	extra output level
`voices`	6	number of voices (1–10)
`voice_mode`	1	1 = Poly I, 2 = Poly II, 0 = Unison
`velocity`	off	velocity sensitivity
`bender_dco / bender_vcf`	0.25 / 0.0	bender depths (0–1)
`mod_depth`	0.5	MOD depth
`master_volume`	0.5	volume
`start_patch`	0	patch at the very first start (afterwards the last one is remembered)
`key_channel`	omni	keyboard channel (omni or 1–16)
`pad_channel`	10	pad channel
`knob_cc`	70 … 77	CC numbers for K1–K8
`pad_notes_a / pad_notes_b`	36 … 43 / 44 … 51	pad notes for bank A / bank B
`mod_cc`	1	CCs that drive MOD (several may be listed)
`knob_mode`	absolute	or relative
`knob_takeover`	jump	or pickup
`save_hold_seconds`	1.5	ignored in 1.2.0; the panel does not write patches
`pad_leds`	on	on = MPK pad LEDs, any = every MIDI out except MIDI 2, off = none

If your MPK sends different numbers

AKAI does not publish a factory-default table for the MPK mini Plus. JU-6 Pi assumes the MPK family's usual defaults: knobs CC 70–77, pads on MIDI channel 10 with notes 36–43 (bank A) and 44–51 (bank B), mod wheel CC 1, sustain CC 64. *NOT VERIFIED on the hardware for this manual.* If your unit differs, either

- change `knob_cc`, `pad_notes_a`, `pad_notes_b` and `pad_channel` in `ju6.conf`, or
- edit the MPK program (Shift + Prog Select, or AKAI's Program Editor) so the pads send those notes and the knobs those CCs in **Absolute** mode.

The three partitions

Partition	Contents	Mounted
bootfs (FAT)	firmware, kernel, config.txt, cmdline.txt	read-only
rootfs (ext4)	Raspberry Pi OS Lite (arm64, Debian 13 "trixie", 2026-10-06) + JU-6 Pi	read-only (overlayroot, changes in RAM)
JU6DATA (FAT, 256 MB)	ju6/ju6.conf, patches.syx, names.txt, state.txt, status.txt	read/write

13 MIDI Implementation

Message	Channel	Function
Note On / Off	omni (or `key_channel`)	play; velocity used only when VELOCITY is on
Notes 36 and 44	10	pad 1, both banks: previous sound (−1, wraps)
Notes 37 and 45	10	pad 2: SIDE 2. LED velocity 127 when that side is active, otherwise 0. Sent for both banks
Notes 38 and 46	10	pad 3: SIDE 1 = VCA LEVEL −; SIDE 2 = pulse on/off
Notes 39 and 47	10	pad 4: SIDE 1 = ENV/GATE; SIDE 2 = saw on/off
Notes 40 and 48	10	pad 5, both banks: next sound (+1, wraps)
Notes 41 and 49	10	pad 6: SIDE 1. LED velocity 127 when that side is active, otherwise 0
Notes 42 and 50	10	pad 7: SIDE 1 = VCA LEVEL +; SIDE 2 = VCF polarity
Notes 43 and 51	10	pad 8: SIDE 1 = CHORUS cycle; SIDE 2 = PWM LFO/MAN
Pitch Bend	any	BENDER (DCO and VCF depth)
CC 1 (and other `mod_cc`)	any	MOD (LFO depth on the DCO)
CC 7	any	master volume
CC 64	any	HOLD (≥ 64 = on)
CC 70–77	any	knobs K1–K8, current side
CC 120, 123	any	all notes off
CC 121	any	reset bender and MOD
Program Change 0–127	any	select A11 … B88
SysEx `F0 41 30 0n pp … F7`	—	Juno-106 patch (APR) into the current sound
SysEx `F0 41 32 0n cc vv F7`	—	Juno-106 single parameter (IPR)

Pad notes are recognised only on the pad channel; notes on other channels (and pad-channel notes that are not pad notes) play the synth. With `pad_leds = on` (the default) JU-6 Pi transmits Note On/Off for every pad on both banks, so the LEDs match the side whichever bank the MPK is showing. Velocity 0 is Note Off. An echo of those notes is ignored for about 80 ms. NOT VERIFIED on an MPK mini Plus.

Patch data format

Each patch is 18 bytes, as in the Juno-106 APR message: 16 slider values (0–127) in the order LFO RATE, LFO DELAY, DCO LFO, DCO PWM, DCO NOISE, VCF FREQ, VCF RES, VCF ENV, VCF LFO, VCF KYBD, VCA LEVEL, ENV A, ENV D, ENV S, ENV R, DCO SUB, followed by two switch bytes:

- **SW1:** bits 0–2 = range 16'/8'/4'; bit 3 = pulse; bit 4 = saw; bit 5 = chorus off (0 = on); bit 6 = chorus I (0 = II).
- **SW2:** bit 0 = PWM MAN (0 = LFO); bit 1 = VCF ENV inverted; bit 2 = VCA GATE (0 = ENV); bits 3–4 = HPF (3 − position).

14 Parameter Chart

Parameter	Side	Control	MIDI	Range / positions	Stored in patch
ENV A	1	K1	CC 70	0–127	slider 12
ENV D	1	K2	CC 71	0–127	slider 13
ENV S	1	K3	CC 72	0–127	slider 14
ENV R	1	K4	CC 73	0–127	slider 15
DCO SUB	1	K5	CC 74	0 = off, 1–127	slider 16
DCO NOISE	1	K6	CC 75	0–127	slider 5
VCF FREQ	1	K7	CC 76	0–127	slider 6
VCF RES	1	K8	CC 77	0–127	slider 7
VCA LEVEL +/−	1	pads 7 / 3	notes 50 / 46	steps of 8, 0–127	slider 11
CHORUS	1	pad 8	note 51	off · I · II · I+II	SW1 b5, b6 (+ ext b0)
VCA ENV/GATE	1	pad 4	note 47	ENV · GATE	SW2 b2
NEXT / PREVIOUS	both	pads 5 / 1	notes 48 / 44	one patch, wraps 0–127	selects a memory
DCO LFO	2	K1	CC 70	0–127	slider 3
DCO PWM	2	K2	CC 71	0–127	slider 4 (0–105 in 106 format)
LFO RATE	2	K3	CC 72	0–127	slider 1
LFO DELAY	2	K4	CC 73	0–127	slider 2
HPF	2	K5	CC 74	0 · 1 · 2 · 3	SW2 b3–4
VCF ENV	2	K6	CC 75	0–127	slider 8
VCF LFO	2	K7	CC 76	0–127	slider 9
VCF KYBD	2	K8	CC 77	0–127	slider 10
PULSE	2	pad 3	note 46	on · off	SW1 b3
SAW	2	pad 4	note 47	on · off	SW1 b4
PWM LFO/MAN	2	pad 8	note 51	LFO · MAN	SW2 b0 (+ ext b1 for ENV)
VCF polarity	2	pad 7	note 50	on = inverted, off = normal	SW2 b1
DCO RANGE	—	—	—	16' · 8' · 4', from the patch	SW1 b0–2
SIDE 1 / SIDE 2	both	pad 6 / pad 2	notes 49 / 45	active pad strong red, the other off	not stored

Side = the panel selected with bank B pad 6 (SIDE 1) or pad 2 (SIDE 2). Slider numbers refer to the 16 slider bytes of the patch data (section 13). "ext" = JU-6 Pi extension column in `names.txt`. Bender depth, MOD depth, master volume, voice mode, velocity and portamento are not on the panel; their start values come from `ju6.conf` and they are not written with a patch.

Patch sheet

Patch	SIDE 1 knobs: A · D · S · R · SUB · NOISE · FREQ · RES							
	SIDE 2 knobs: DCO LFO · PWM · LFO RATE · DELAY · HPF · VCF ENV · VCF LFO · KYBD							
	Pads: NEXT · PREV · LEVEL · CHORUS · ENV/GATE · PULSE · SAW · LFO/MAN · POLARITY							

Copy this sheet to note down your own sounds.

15 Patch List

Banks A and B contain the Juno-106 factory patches as provided by the KR-106 project (parameter values, not samples), in exact 106 format (`patches/factory_106.syx`). Columns are the patch number inside the group. Reach a row by stepping one sound at a time (section 7) or with Program Change.

Bank A	1	2	3	4	5	6	7	8
A1x	Brass	Brass Swell	Trumpet	Flutes	Moving Strings	Brass & Strings	Choir	Piano I
A2x	Organ I	Organ II	Combo Organ	Calliope	Donald Pluck	Celeste	E Piano I	E Piano II
A3x	Clock Chimes	Steel Drums	Xylophone	Brass III	Fanfare	Strings III	Pizzicato	High Strings
A4x	Bass Clarinet	English Horn	Brass Ensemble	Guitar	Koto	Dark Pluck	Funky I	Synth Bass I
A5x	Lead I	Lead II	Lead III	Funky II	Synth Bass II	Funky III	Thud Wah	Going Up
A6x	Piano II	Clav.	Frontier Organ	Snare Drum	Tom Toms	Timpani	Shaker	Synth Pad
A7x	Sweep I	Pluck Sweep	Repeater	Sweep II	Pluck Bell	Dark Synth Piano	Sustainer	Wah Release
A8x	Gong	Resonance Funk	Drum Booms	Dust Storm	Rocket Men	Hand Claps	FX Sweep	Caverns

Bank B	1	2	3	4	5	6	7	8
B1x	Strings	Violin	Chorus Vibes	Organ I	Harpsichord I	Recorder	Perc Pluck	Noise Sweep
B2x	Space Chimes	Nylon Guitar	Orchestral Pad	Bright Pluck	Organ Bell	Accordion	FX Rise 1	FX Rise 2
B3x	Brass	Helicopter	Lute	Chorus Funk	Tomita	FX Sweep 1	Sharp Reed	Bass Pluck
B4x	Resonant Rise	Harpsichord 2	Dark Ensemble	Contact Wah	Noise Sweep 2	Glassy Wah	Phase Ensemble	Chorused Bell
B5x	Clav	Organ 2	Bassoon	Auto Noise Sweep	Brass Ensemble	Ethereal	Chorus Bell 2	Blizzard
B6x	E Piano Trem	Clarinet	Thunder	Reedy Organ	Flute / Horn	Toy Rhodes	Surf's Up	OW Bass
B7x	Piccolo	Melodic Taps	Meow Bass	Violin (high)	High Bells	Rolling Wah	Ping Bell	Brassy Organ
B8x	Low Dark Strings	Piccolo Trumpet	Cello	High Strings	Rocket Men	Forbidden Planet	Froggy	Owgan

Names from `src/factory_bank.h`. Memories you write are renamed "*xNN Egen*".

16 Troubleshooting

Symptom	Check
No ready tones, LED keeps blinking slowly	No audio output found. With HDMI, use **HDMI0** (nearest USB-C). Check the receiver's input. Read `ju6/status.txt`.
Ready tones fall (E–A) instead of rising	`audio_output = dacplus` but the DAC+ was not found — check that it is seated correctly. Output is on HDMI/USB meanwhile.
LED on with a short gap every 2 s	No MIDI device: check the MPK's USB cable. Devices can be plugged in while running.
Keys play but knobs/pads do nothing or do the wrong thing	The MPK program sends other CC/note numbers or another pad channel. Adjust `ju6.conf` or the MPK program (section 12). Make sure the pads are in the bank you expect (BANK A/B).
Pad presses play notes	Pads not on the pad channel (default 10) — set `pad_channel` to match.
Knobs have no effect until turned past a point	`knob_takeover = pickup` is set — this is soft takeover.
Noticeable delay with a TV or AV receiver	Select Game/Direct/Pure mode, or use a DAC+ or HDMI audio extractor.
Clicks or drop-outs	The buffer grows automatically after repeated underruns; you can also set `buffer = 512x2`.
The octave is wrong for this patch	DCO RANGE is stored in the patch and is not on the panel. Step to another sound, or load a SysEx patch with the range you want. OCT −/+ only shifts the MPK keyboard.
Notes hang	Release the sustain pedal, or send CC 123 (all notes off).
Pad LEDs do not show the side	Set `pad_leds = on` (default). The MPK must appear as a USB-MIDI port with “MPK” in the name. NOT VERIFIED on hardware; the Pi LED still blinks once for SIDE 1 and twice for SIDE 2. `pad_leds = off` stops the messages.
An edited sound was not kept after writing	Fixed in v1.1.1 (section 10); update the SD card.
Want factory sounds back	Delete `patches.syx` and `names.txt` on JU6DATA.

With a screen and keyboard on the Pi

`ju6pi list` (patch names), `ju6pi bench` (CPU margin), `ju6pi demo DIR` (render WAV examples), `ju6pi utganger` (sound cards and chosen output), `ju6pi selftest` (MPK mapping, sound step, SysEx, output selection), `journalctl -u ju6pi` (log). `aplay -l` should list `RPiDAC [RPi DAC+]` when the DAC+ is fitted.

17 Specifications

Sound engine	Ultramaster KR-106 DSP (component-level Juno-106 model), GPL-3, unmodified in third_party/kr106
Polyphony	6 voices (configurable 1–10); Poly I, Poly II, Unison
DCO	Pulse with PWM (LFO / MAN / ENV), sawtooth, sub-oscillator (square, −1 oct), noise; range 16' / 8' / 4'
VCF	4-pole resonant low-pass (IR3109 character), self-oscillating; ENV ±, LFO, KYBD; 2× or 4× oversampling
HPF	4 positions: 0 bass boost (≈ +9.4 dB shelf), 1 flat, 2 ≈ 236 Hz, 3 ≈ 754 Hz
VCA	ENV or GATE; LEVEL ≈ −10 … +10 dB
ENV / LFO	One ADSR (Juno-106 curves); triangle LFO with RATE and DELAY, AUTO or MANUAL trigger
Chorus	BBD model (MN3009 type), modes off / I / II / I+II, stereo
Memory	128 patches (A11–B88), Juno-106 SysEx (APR) format; factory banks A and B
Controller	AKAI MPK mini Plus over USB (any class-compliant USB-MIDI device, hot-plug)
Audio outputs	Raspberry Pi DAC+ (RCA + 3.5 mm, 48 kHz/32-bit) · HDMI0 / HDMI1 (2-ch PCM, 48 kHz) · USB audio
Sample rate	48 kHz
Buffer at start	DAC+ 5.3 ms · HDMI 10.7 ms · USB 2.7 ms (automatically increased on underruns)
Latency	≈ 8–10 ms (DAC+), ≈ 12–17 ms (HDMI) — calculated, not measured
Start-up time	≈ 10–15 s to ready tones
Hardware	Raspberry Pi 5, 8 GB; microSD ≥ 8 GB; 27 W USB-C supply recommended
Operating system	Raspberry Pi OS Lite arm64 (Debian 13 “trixie”), read-only root (overlayroot)
Software	ju6pi 1.2.0 (9 November 2026), C++17, ALSA; SCHED_FIFO 80 on isolated CPU 3
Licence	GPL-3.0

Specifications are taken from the v1.2.0 source and README and are subject to change.

NOTE Chorus II and I+II speeds have not yet been verified against hardware by the KR-106 project; the chorus circuit is modelled on the Juno-6/60 family, which shares its topology with the 106. Analogue voice-to-voice variation is only approximated.

18 Sources and Credits

What this manual is based on

Topic	Source (repository OleBee/juno106-pi, v1.2.0)
Two sides, pad functions, page LEDs, next/previous sound, Program Change	`src/controller.h`, `src/midi_out.cpp`
Parameter scaling, patch format, MOD/bender, transpose key	`src/synth.h`, `third_party/kr106/DSP/KR106SysEx.h`, `KR106_DSP_SetParam.h`, `KR106_HPF.h`, `KR106LFO.h`
Defaults, LED, ready tones, status file, oversampling, buffer step-up	`src/main.cpp`, `image/data/ju6.conf`
Audio outputs, DAC+ detection, buffer tables	`src/audio_out.h`, `image/boot/dacplus.blokk`
Patch memory and files	`src/bank.h`, `src/factory_bank.h`, `image/data/LES-MEG.txt`
Installation, latency estimates, MPK defaults, changelog	`README.md`
MPK mini Plus layout and controls	AKAI Professional, *MPK mini Plus User Guide v1.2* (diagram redrawn)

Credits

- Sound engine: **Ultramaster KR-106** by Karl LaRocca and the original Ultramaster group (GPL-3).
- Raspberry Pi OS © Raspberry Pi Ltd. and the Debian project.
- AKAI Professional and MPK are trademarks of inMusic Brands. Diagrams in this manual are original drawings.
- The layout of this manual is inspired by 1980s synthesizer owner's manuals; no text has been copied.
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